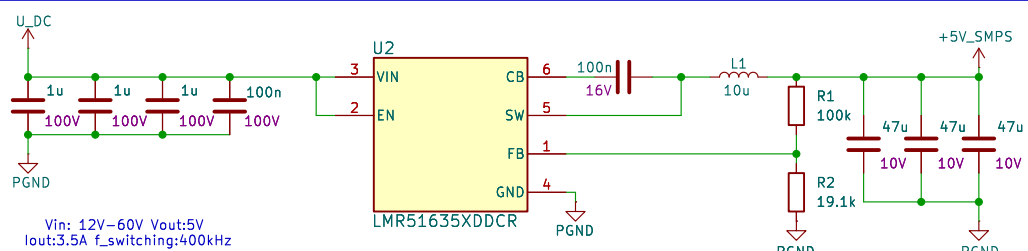






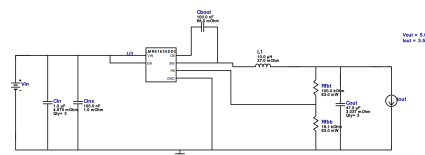
# Buck Converter



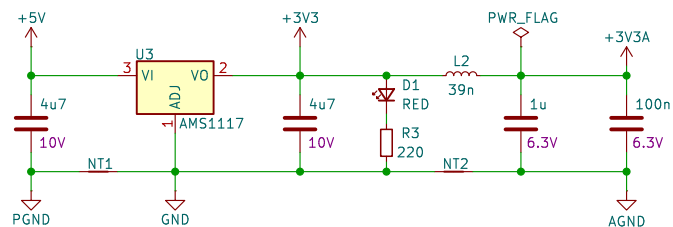
TEXAS  
INSTRUMENTS  
WEBENCH® Design Report

Design: 120 LMR51635DDC  
LMR51635DDC 12V-60V to 5.0V @ 3.5A

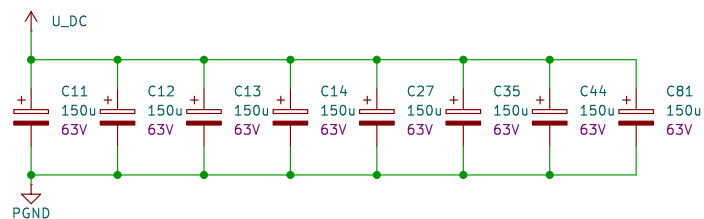
VinMin = 12.0V  
VinMax = 60.0V  
Vout = 5.0V  
Iout = 3.5A  
Device = LMR51635DDC  
Topology = Buck  
Created = 2020-01-18 17:57:50.631  
BOM Count = 12  
Total PS = 2.47W



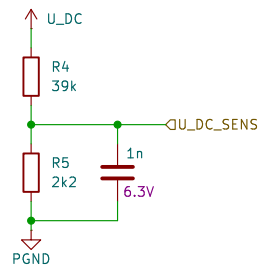
# Low drop-out regulator & GNDA / GND separation



# Bulk Capacitors



# DC-Link Voltage



Sheet: /Power/  
File: power.kicad\_sch

Title:

Size: A4

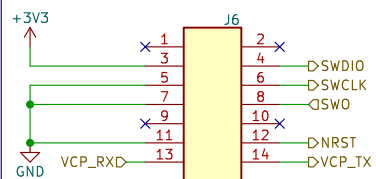
Date:

KiCad E.D.A. 9.0.7

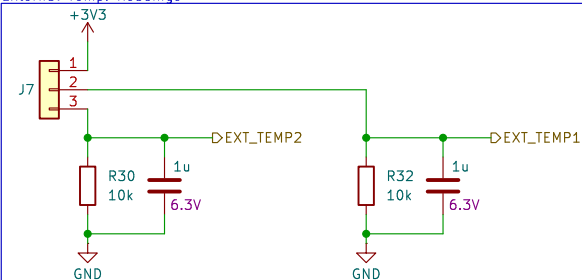
Rev:

Id: 3/7

ST-Link V3 MINIE CONNECTOR (STDC14)

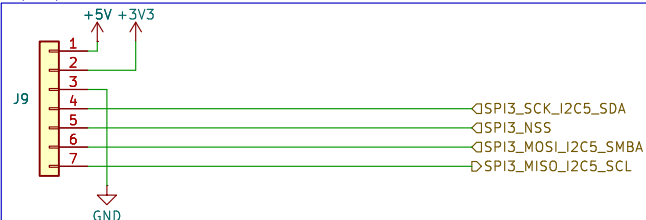


External Temp. Readings

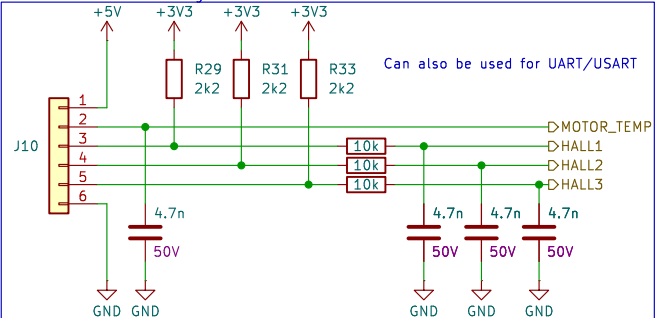


for motor and etc.

SPI/I2C/USART connection



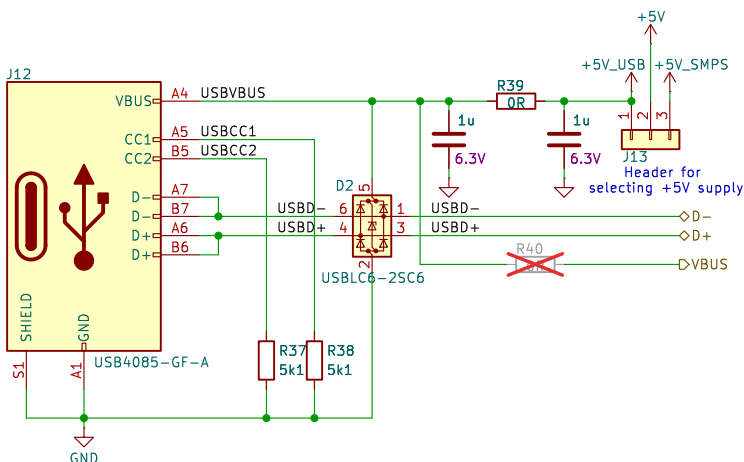
Hall connector and filtering



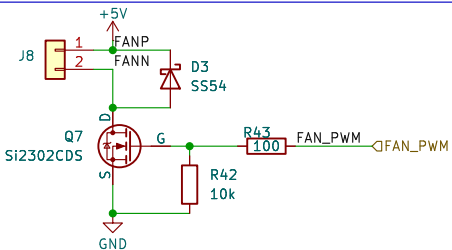
CAN Connector



USB-C Connector



Fan connector



todo: check JST connectors for pitch and consider changing them to SMD and include it in assembly.

Sheet: /Connectors/  
File: connectors.kicad\_sch**Title:**

Size: A4

Date:

KiCad E.D.A. 9.0.7

**Rev:**

Id: 4/7





